



Ts. Dr. Alawi Alqushaibi

Program Coordinator - Information Technology · Lecturer in Artificial Intelligence

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[+601121226508](tel:+601121226508)

alqushaibi.github.io

[Google Scholar](#)

[MAHSA University, Malaysia](#)

8+ EXPERIENCE

2,441 CITATIONS

18 H-INDEX

23 I10-INDEX

PROFESSIONAL SUMMARY

Program Coordinator and Lecturer in Artificial Intelligence at MAHSA University with expertise in Artificial Intelligence, Data Science, Digital Technologies, and Applied Machine Learning. Over eight years of academic and industry experience spanning higher education teaching, curriculum development, accreditation support, postgraduate supervision, and interdisciplinary AI research. My academic work integrates technology innovation with practical industry applications including healthcare AI, intelligent systems, computer vision, optimization algorithms, and digital transformation initiatives. I have established a strong research profile with publications in high-impact Q1/Q2 journals, more than 2,200 citations, and active collaborations with industry and research institutions. I am committed to student-centred education, industry-engaged learning, and research-led teaching aligned with international higher education standards.

EDUCATION

Ph.D. in Information Technology

Universiti Teknologi PETRONAS, Malaysia

Jan 2022 - May 2025

Optimization of Pix2pix GANs via Gradient Penalty Sine Cosine Algorithm

M.S. in IT, by Research

Universiti Teknologi PETRONAS, Malaysia

Jul 2019 - Aug 2021

Enhanced Weight Optimized RNN Based on Sine Cosine Algorithm

B.S. in Computer Science

Universiti Teknologi Malaysia

Jul 2008 - Mar 2012

Computer Network and Security - CGPA 3.10/4 (2nd Upper)

AWS re/Start Program

Amazon Web Services (AWS) Training Program

Dec 2025 - Mar 2026

Cloud Architecture, DevOps, Security, AI/ML on AWS

RESEARCH INTERESTS

Machine Learning Deep Learning
Artificial Intelligence Generative AI
Image Processing Data Science
Optimization Metaheuristics
Computer Vision Medical Imaging GANs
Reinforcement Learning

TECHNICAL SKILLS

MATLAB Proficient
Python Advanced
SQL Proficient
Linux Advanced
AWS Cloud Advanced

PROFESSIONAL EXPERIENCE

Program Coordinator: Information Technology

MAHSA University Current

Jan 2026 - Present

- Coordinate academic and administrative operations of the IT program
- Oversee curriculum planning, course mapping, and syllabus alignment with MQF and accreditation standards
- Coordinate course offerings, teaching assignments, and semester scheduling
- Liaise with faculty members, Dean/Deputy, and university management on program matters
- Monitor course delivery quality and learning outcomes achievement
- Support accreditation, audit, and quality assurance documentation
- Handle student academic issues, progression matters, and program-level advising
- Review and improve assessment methods and moderation processes
- Assist in strategic planning and continuous improvement initiatives for the IT program

Lecturer in Artificial Intelligence

MAHSA University Current

Jul 2025 - Present

- IT Lecturer, Faculty of Engineering, Built Environment and Information Technology, MAHSA University
- Teach undergraduate courses: Data Science, Artificial Intelligence, Wireless Communication, IT Fundamentals, Operating Systems, DBMS, Information Management, Information System, Introduction to Programming, OOP, and Computer Architecture & Organization
- Design, prepare, and update lecture materials, tutorials, assignments, and assessments
- Design, prepare, and teach Standalone Micro Credential Programs
- Supervise PhD students, co-supervise PhD students, co-supervisor postdoctoral projects, and supervise FYP students
- Conduct lectures, tutorials, and practical sessions using active and student-centered learning approaches
- Evaluate student performance through continuous assessment and final examinations
- Participate in curriculum development and course review
- Engage in academic research and publish findings in peer-reviewed journals and conferences
- Provide academic advising and mentoring to students
- Support institutional activities, accreditation, and quality assurance processes

LANGUAGES

Arabic Native



English Fluent



AWARDS & RECOGNITION

Graduate on Time (GOT) Award- PhD

Universiti Teknologi PETRONAS (UTP) · 2025

For completion of PhD in the stipulated time of 42 months

Graduate on Time (GOT) Award - M.S.

Universiti Teknologi PETRONAS (UTP) · 2021

For completion of M.S. in the stipulated time of 24 months

Best Paper Award - International Conference

Pimpri Chinchwad College of Engineering, Pune, India · 2024

Recognition for outstanding research paper

Publication Achievement Recognition

Universiti Teknologi PETRONAS (UTP) · 2024

For outstanding publication record during doctoral studies

International Students Leadership Conference

Ministry of Higher Education Malaysia · 2010

Yemen Scholarship for Outstanding Students

Ministry of Higher Education, Yemen · 2007

Awarded for bachelor's degree studies abroad

PROFESSIONAL MEMBERSHIPS

Professional Technologist (Ts.) — Malaysia Board of Technologists (MBOT)

Graduate Technologist — Malaysia Board of Technologists (MBOT)

Centre for Research in Data Science (CERDAS) — Universiti Teknologi PETRONAS

International Association of Engineers (IAENG) — Member

Research Fellowship — Research Management Centre (RMC), MAHSA University

JOURNAL REVIEWER

Computers, Materials & Continua

IEEE Access

PLOS ONE

Computer Systems Science & Engineering

Franklin Open

International Journal of Computer Mathematics

WORKSHOPS & TRAINING

Outcome-Based Education (OBE) / Malaysian Qualifications Framework (MQF) Expert Assessor

MAHSA University

Current

April 2026 - Present

- a) Programme Rationale and Objectives
- b) Programme Learning Outcomes (aligned with MQF Level 8)
- c) Curriculum Design and Structure
- d) Delivery Methods for ODL, including Self-Instructional Materials (SIM)
- e) Assessment Strategies suitable for remote learning
- f) Academic and Learner Support Services
- g) Facilities and Learning Resources for online learners
- h) Quality Assurance and Programme Monitoring Mechanisms

Lecturer of Computer Science

Yemeni Military College

6 Years

Jul 2012 - May 2018

- Plan, prepare and conduct C++ and Java lectures
- Apply interactive teaching methods to boost engagement
- Undergraduate lab demonstration and CS lecturing
- Supervised and mentored Bachelor's degree students
- Facilitated hands-on technical workshops and labs

Information Technology Support

ML Company for Training and Logistic Services

1 Year

Jun 2018 - Jun 2019

- Preparing and writing daily reports and financial support administration
- Hard and soft copy archiving for company and project documents
- Plan and manage timelines to ensure goals are met on time
- Preparing and translating course slides (English to Arabic) and crane user manuals
- Communicating with port departments and providing comprehensive logistical support

System Administrator Assistant (Part Time)

Sabafon Company, Yemen

6 Years

Jun 2012 - May 2018

- Install, manage and upgrade Windows Server 2008/2012/2016 and apply patches
- Create domain users and manage existing user properties
- Manage hardware installation and maintenance of Intel servers (Disk, CPU/Memory, PCI, HBA, SAS)
- Manage and configure Windows DNS/DHCP service
- Physical installation of server racks, KVM, switches and rack mount keyboards
- LAN, WAN, WLAN – Installation, Configuration, and Troubleshooting

Practical Training (Bachelor Degree)

Ecosensa Technologies Sdn. Bhd, Malaysia

6 Months

Jul 2011 - Dec 2011

- Installing RFID devices, configuring routers, and troubleshooting gantry problems
- Requirements gathering, development planning, designing, workshops, documentation, and MySQL database maintenance
- Promote company products and services and convince customers to sign contracts

PUBLICATIONS

15 TOTAL · 11 JOURNALS · 3 CONFERENCES · 1 BOOK CHAPTERS

Journal Articles

[1]

RNN-LSTM: From applications to modeling techniques and beyond—Systematic review

SM Al-Selwi, MF Hassan, SJ Abdulkadir, A Muneer, EH Sumiea, **A Alqushaibi**, MG Ragab
Journal of King Saud University-Computer and Information Sciences 36(5), 2024

Q1

IF 5.2

ISI, Scopus

655

Student-Centered Learning: Engaging Learners through Active Learning

Universiti Teknologi PETRONAS · 2024

Computer Vision with MATLAB

Universiti Teknologi PETRONAS · 2022

Introduction to Artificial Intelligence

Yemeni Student Union: Malaysia · 2020

How To Be A Data Scientist

Yemeni Student Union : Malaysia · 2020

AI-100T01: Designing & Implementing an Azure AI Solution (3 days)

Universiti Teknologi PETRONAS · 2020

Diploma of Maintenance and Programming Mobile

Science and Life Institution · 2018

Professional Skills

Universiti Teknologi Malaysia · 2012

International Student Leaders Conference 2010 (7 days)

Ministry of Higher Education Malaysia · 2010

ACADEMIC LEADERSHIP & QUALITY ASSURANCE

- Program coordination
- MQA/MQF compliance
- Curriculum mapping
- CLO/PLO alignment
- Assessment moderation
- Continuous Quality Improvement (CQI)
- Accreditation preparation
- Student advising
- Timetable coordination
- Teaching allocation

[2]

Enhanced weight-optimized recurrent neural networks based on sine cosine algorithm for wave height prediction

A Alqushaibi, SJ Abdulkadir, HM Rais, Q Al-Tashi, MG Ragab, H Alhussian
Journal of Marine Science and Engineering 9(5), 2021

Q2 IF 2.458 ISI, Scopus 33

[3]

Smart grid stability prediction using adaptive aquila optimizer and ensemble stacked BiLSTM

SM Al-Selwi, MF Hassan, SJ Abdulkadir, MG Ragab, **A Alqushaibi**, EH Sumiea
Results in Engineering 24, 2024

Q1 IF 3.4 Scopus 22

[4]

Enhanced colon cancer segmentation and image synthesis through advanced generative adversarial networks based-Sine Cosine Algorithm

A Alqushaibi, MH Hasan, SJ Abdulkadir, KU Danyaro, MG Ragab, SM Al-Selwi, EH Sumiea, H Alhussian
IEEE Access 12, 2024

Q1 IF 3.4 ISI, Scopus 20

[5]

Type 2 diabetes risk prediction using deep convolutional neural network based-Bayesian optimization

A Alqushaibi, MH Hasan, SJ Abdulkadir, A Muneer, M Gamal, Q Al-Tashi, SM Taib, H Alhussian
Computers, Materials & Continua 75(2), 2023

Q2 IF 3.860 ISI, Scopus 18

[6]

Segmentation of head and neck tumors using dual PET/CT imaging: UNet transformer-based comparative analysis of 2D, 2.5D, and 3D approaches

MA Mahdi, S Ahamad, SA Saad, A Dafhalla, **A Alqushaibi**, R Qureshi
Computer Modeling in Engineering & Sciences 2024

Q2 Scopus 5

[7]

A comprehensive systematic review of YOLO for medical object detection (2018 to 2023)

MG Ragab, SJ Abdulkadir, A Muneer, **A Alqushaibi**, EH Sumiea, R Qureshi, SM Al-Selwi, H Alhussian
IEEE Access 12, 2024

Q1 IF 3.4 ISI, Scopus 4

[8]

A systematic review of material erosion prediction techniques: Incorporating model parameters variability and the lack of field-scale representation

A Alghurabi, A Abduljabbar, MA Maoinsar, **A Alqushaibi**, H Hagar, ME Mohyaldinn
Results in Engineering 2025

Q1 IF 3.4 Scopus 4

[9]

Periodontitis bone loss detection in panoramic radiographs using modified YOLOv7

MG Ragab, SJ Abdulkadir, N Qaid, TM Gondal, **A Alqushaibi**, R Qureshi, F Shaukat
PeerJ Computer Science e3102, 2025

Q2 Scopus 3

[10]

Enhancing predictive accuracy for recurrence-free survival in head and neck tumor: A comparative study of weighted fusion radiomic analysis

MA Mahdi, S Ahamad, SA Saad, A Dafhalla, **A Alqushaibi**, R Qureshi
Diagnostics 2024

Q2 Scopus 3

[11]

The future of green transportation: A review of technological advancements in electric powered vehicles

H Shutari, NS Ahmad, YZ Alharthi, **A Alqushaibi**, AS Algamili
IETE Technical Review 2026

Q2 ISI, Scopus 1

Conference Papers

[1]

An optimized recurrent neural network for metocean forecasting

A Alqushaibi, SJ Abdulkadir, HM Rais, Q Al-Tashi, MG Ragab

2020 International Conference on Computational Intelligence (ICCI) pp. 190-195, 2020

Conference

ISI, Scopus

99 4

[2]

A comparative analysis of machine learning techniques for heart disease prediction

A Alqushaibi, MG Ragab, EH Sumiea, SM Al-Selwi, MH Hasan, SJ Abdulkadir, H Alhussian

2023 IEEE 21st Student Conference on Research and Development (SCoReD) pp. 579-584, 2023

Conference

ISI, Scopus

99 3

[3]

Brain tumor detection instance segmentation using YOLOv8 in medical imaging

MG Ragab, KU Danyaro, SJ Abdulkadir, SM Al-Selwi, EH Sumiea, **A Alqushaibi**

2024 IEEE Asia-Pacific Conference on Applied Electromagnetics (APACE) pp. 55-58, 2024

Conference

IEEE, Scopus

99 3

Book Chapters

[1]

Moth-flame optimization algorithm for feature selection: A review and future trends

Q Al-Tashi, S Mirjalili, J Wu, SJ Abdulkadir, TM Shami, N Khodadadi, **A Alqushaibi**

Handbook of Moth-Flame Optimization Algorithm, CRC Press pp. 11-34, 2022

Book Chapter

Scopus

99 25

RESEARCH PROJECTS

Collaborative AI research initiative with Malaysian Nuclear Agency involving deep learning-based material classification for intelligent scanning systems

[Active : 2026-Present](#)

Deep learning framework for nuclear scanning equipment: material classification (gas, liquid, air, solid). Collaboration with the Malaysian Nuclear Agency.

Deep Learning Teeth Age Prediction from Dental Radiographs · [Active : 2026-Present](#)

AI model for biological age estimation from dental X-rays. Cross-faculty collaboration with MAHSA Medic (IT & Dentistry).

FOEBEIT Academic Planner · [Active : 2025-Present](#)

Timetable Generation and Human Resource Planning System for the Faculty of Engineering, Built Environment and IT :MAHSA University.

ConfTrack Malaysia Conference Management System · [Active : 2026 -Present](#)

Full-lifecycle platform: paper submission, blind peer review, editorial decisions, QR attendance, badges, and digital certificates. Hosting MISEBEIT 2026.
www.conftrack.com

Autonomous Underwater Defect Detection · [Active: 2023-Present](#)

CNN-based detection of structural defects (corrosion, cracks, biofilm) in offshore oil & gas platforms. Collaboration with UTP.

Head & Neck Tumor Segmentation in PET-CT · [Completed: 2023-2025](#)

Transformer-based segmentation for radiotherapy planning. Published in IEEE Access, Diagnostics, and CMES. (Project ID: RG-23137)

Scalable Automated Data Preprocessing Framework · [Completed: 2022-2024](#)

For Very Big Data Stream in Oil Platform Equipment's Time to Failure Prediction Model: UTP.

Supervised ML for Offshore Defect Mapping · [Completed: 2022-2024](#)

Fundamental study of supervised machine learning techniques for autonomous defect mapping of offshore structures: UTP.

Digital Twin for Structural Asset Monitoring · [Completed: 2019-2021](#)

Solution and decision making for onshore facilities: UTP.

Resultant impact assessment on offshore structures loading: UTP.

TEACHING

Artificial Intelligence - Micro Credential Undergraduate · ai-data Currently Teaching	Data Science - Micro Credential Undergraduate · ai-data Currently Teaching
Introduction to Programming Undergraduate · programming Currently Teaching	Object-Oriented Programming Undergraduate · programming Past
Information Technology Fundamentals Undergraduate · info-systems Past	Database Management Systems Undergraduate · info-systems Currently Teaching
Information Management Undergraduate · info-systems Past	Information System Undergraduate · info-systems Currently Teaching
Wireless Communication Undergraduate · networks-systems Past	Operating Systems Undergraduate · networks-systems Past
Computer Architecture & Organization Undergraduate · networks-systems Past	Mobile Technology Undergraduate · networks-systems Currently Teaching
TCP/IP Programming Undergraduate · networks-systems Currently Teaching	

STUDENT SUPERVISION

PhD & Master Students					
Name	Degree	University	Role	Research Area	Status
Raju Aparadhi	Ph.D. Candidate	MAHSA University	Main Supervisor	AI / Machine Learning	currently
Bashir Ibne Zafor	Ph.D. Candidate	MAHSA University	Main Supervisor	AI, Machine Learning	currently
Xu Heng	Ph.D. Candidate	MAHSA University	Co-Supervisor	Intelligent Construction	currently
Final Year Projects (FYP)					
Student	Project Title			Role	Status
Mustafa Mohamed	Sudan Donation Website			Main Supervisor	completed

Student	Project Title	Role	Status
Mutaman Fadhil	Design and Implementation of a Tourist Website	Main Supervisor	completed
Mohannad Algaali Mohammed Abdullah	Smart Waste Management System Using Image Recognition	Main Supervisor	current
Md Abdul Kader Jahid	Intelligent Chatbot System for Student Enquiry Support Using Natural Language Processing	Main Supervisor	current
Mohamad Adham Ikram Bin Ideris	Email Phishing Detection System using Natural Language Processing (NLP)	Main Supervisor	current

LEADERSHIP & VOLUNTEER

President: Association of International Students (Yemen Chapter)

Universiti Teknologi PETRONAS, Malaysia

2023 - 2025

- Represented the chapter in all internal and external matters; led executive committee meetings and activities
- Served as main liaison between the Yemen Chapter and the main association body and university
- Set short-term and long-term goals; oversaw planning and implementation of annual programs and cultural events
- Supervised and supported all executive members; delegated responsibilities and ensured deadlines are met
- Addressed academic, social, and cultural concerns of Yemeni students; provided guidance to new arrivals
- Advocated for inclusion, diversity, and student welfare within the university

Vice President : Yemeni Student Society

Universiti Teknologi Malaysia (UTM)

Jan 2009 - Mar 2010

- Co-established the Yemeni Student Society at UTM
- Managed regular meetings, planned activities and trips for society members
- Solved student problems and followed up with concerned university departments

Member : AIESEC

AIESEC in Universiti Teknologi Malaysia

May 2010 - May 2012

- Social service and conference preparation
- Hosted international guests, arranged airport pickup and accommodation in Malaysia
- Educated people about HIV/AIDS prevention in lectures and community outreach
- Worked in special needs schools helping disabled students

REFEREES

Assoc. Prof. Ts. Dr Mohd Hilmi Hasan

Chair, Computer & Information Sciences Department, Universiti Teknologi PETRONAS

✉ mhilmi_hasan@utp.edu.my

📍 Universiti Teknologi PETRONAS, 32610 Seri Iskandar, Perak, Malaysia

Relationship: PhD. Supervisor

Ts. Vickneswari A/P Durairajah

Deputy Dean - Undergraduate, Postgraduate & Research

✉ vickneswari@mahsa.edu.my

📍 Faculty of Engineering, Built Environment & Information Technology, MAHSA University

Dr. Kamaluddeen Usman Danyaro

✉ kamaluddeen.usman@utp.edu.my
📍 Universiti Teknologi PETRONAS,
32610 Seri Iskandar, Perak, Malaysia

Eng. Abdulhadi Saeed Hasan Al-Zawari

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